RubricEye Assessment

Structured marking rubric

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Marking criteria

SECTION B — Short Answer Questions (20 Marks)

2i — 4 marks

State Newton's Second Law of Motion and write its mathematical expression.

- Award marks for: 1) Correctly stating Newton's Second Law (e.g., rate of change of momentum is directly proportional to applied force, or F=ma). 2) Providing the correct mathematical expression ($F = ma$ or $F = dp/dt$).

Source note: State Newton's Second Law of Motion and write its mathematical expression. · high confidence

2ii — 4 marks

Differentiate between RAM and ROM. Give one example of each.

- Award marks for: 1) Differentiating RAM (volatile, read/write, temporary) and ROM (non-volatile, read-only, permanent). 2) Providing one valid example for RAM (e.g., DRAM, SRAM). 3) Providing one valid example for ROM (e.g., PROM, EEPROM, BIOS chip).

Source note: Differentiate between RAM and ROM. Give one example of each. · high confidence

2iii — 4 marks

Explain the role of the nucleus in a cell.

- Award marks for explaining the nucleus's role as the control center of the cell, detailing that it contains genetic material (DNA) and regulates cell activities, growth, and reproduction/gene expression.

Source note: Explain the role of the nucleus in a cell. · high confidence

2iv — 4 marks

What is the difference between a series and a parallel circuit? Also Draw a diagram.

- Award marks for: 1) Differentiating series (single path for current, components share voltage) and parallel (multiple paths, components share current) circuits. 2) Drawing a clear, correctly wired diagram illustrating the difference (either showing both or a combined representation as appropriate).

Source note: What is the difference between a series and a parallel circuit? Also Draw a diagram. · high confidence

2v — 4 marks

Name two input devices and two output devices used in a computer system. And what operation do they perform.

- Award marks for: 1) Naming two correct input devices (e.g., keyboard, mouse, microphone). 2) Naming two correct output devices (e.g., monitor, printer, speakers). 3) Describing the operation each performs (e.g., input devices send data/commands to the CPU; output devices present processed data to the user).

Source note: Name two input devices and two output devices used in a computer system. And what operation do they perform. · high confidence

2vi — 4 marks

A force acts on a box on a horizontal surface. Study the given force diagram and identify the forces acting upward, downward, forward and backward. Complete the diagram with proper labeling.

- Award marks for: 1) Correctly identifying the forces acting upward (e.g., normal force), downward (e.g., gravity/weight), forward (e.g., applied force), and backward (e.g., friction) based on the provided diagram. 2) Completing the diagram with proper, clear labeling of these force vectors.

Source note: identify the forces acting upward, downward, forward and backward. Complete the diagram with proper labeling. · high confidence

2vii — 4 marks

Draw a bar graph representing the following data: Student Marks (Total: 50) Ahmed 45 Sara 20 Ali 35 Khadija 40

- Award marks for: 1) Drawing a bar graph with correctly labeled axes (Student on x-axis, Marks on y-axis). 2) Accurately plotting the bars to scale for Ahmed (45), Sara (20), Ali (35), and Khadija (40) out of a total of 50.

Source note: Draw a bar graph representing the following data: Student Marks (Total: 50) Ahmed 45 Sara 20 Ali 35 Khadija 40 · high confidence

SECTION C — Structured / Extended Response (15 Marks)

3a — 8 marks

Explain the laws of reflection of light. In your answer, define the incident ray, reflected ray, normal, angle of incidence, and angle of reflection. State the relationship between the angle of incidence and the angle of reflection.

- Award marks for: 1) Stating the laws of reflection (incident ray, reflected ray, and normal lie in the same plane; angle of incidence equals angle of reflection). 2) Defining incident ray, reflected ray, normal, angle of incidence, and angle of reflection. 3) Explicitly stating the mathematical relationship ($\angle i = \angle r$).

Source note: Explain the laws of reflection of light. In your answer, define the incident ray, reflected ray, normal, angle of incidence, and angle of reflection. State the relationship between the angle of incidence and the angle of reflection. · high confidence

3b — 7 marks

Draw a complete ray diagram showing the reflection of a light ray from a plane mirror. Your diagram must clearly show and label the plane mirror, incident ray, reflected ray, normal, angle of incidence, and angle of reflection.

- Award marks for drawing a complete and accurate ray diagram showing light reflecting off a plane mirror. The diagram must clearly show and correctly label: the plane mirror, incident ray, reflected ray, normal (dashed line perpendicular to mirror), angle of incidence, and angle of reflection.

Source note: Draw a complete ray diagram showing the reflection of a light ray from a plane mirror. Your diagram must clearly show and label the plane mirror, incident ray, reflected ray, normal, angle of incidence, and angle of reflection. · high confidence

4a — 8 marks

What is an algorithm? Explain any three characteristics of a good algorithm. Also state one advantage of writing an algorithm before solving a programming problem.

- Award marks for: 1) Defining an algorithm (step-by-step procedure to solve a problem). 2) Explaining three characteristics of a good algorithm (e.g., unambiguous, finite, effective, definite input/output). 3) Stating one advantage of writing an algorithm before programming (e.g., serves as a blueprint, easier debugging, breaks down complex problems).

Source note: What is an algorithm? Explain any three characteristics of a good algorithm. Also state one advantage of writing an algorithm before solving a programming problem. · high confidence

4b — 7 marks

Draw a complete flowchart to find and display the largest of three numbers entered by the user. Use the correct standard flowchart symbols, decision branches, and arrows.

- Award marks for: 1) Drawing a complete flowchart that logically finds and displays the largest of three numbers. 2) Using correct standard flowchart symbols (terminals, processes, decisions, input/output). 3) Including correct decision branches (comparisons between the three numbers) and directional arrows. 4) Displaying the final result.

Source note: Draw a complete flowchart to find and display the largest of three numbers entered by the user. Use the correct standard flowchart symbols, decision branches, and arrows. · high confidence